

PHINIA Inc.

AI Research Project · 2026

Fuel Efficiency Predictor

A complete machine learning application that predicts vehicle fuel efficiency and CO₂ emissions across hydrogen, CNG, ethanol, petrol and diesel — designed to support PHINIA's clean energy mission.

Built by Rory Pius · BSc Computer Science (AI), Year 1 · Project Submission for PHINIA Inc., April 2026

392

Vehicles Trained

0.88

Model Accuracy
R²

5

Fuel Types

3

App Layers Built

Live

GitHub Repository

01 — Executive Summary

This document presents the Fuel Efficiency Predictor: a full-stack machine learning application built by Rory Pius as an unsolicited research project demonstrating direct relevance to PHINIA's core business and clean energy mission.

The system predicts vehicle fuel efficiency, CO₂ emissions, and fuel consumption based on engine specifications and fuel type. It uses a Random Forest machine learning model trained on 392 real vehicles from the UCI Auto MPG dataset, achieving an R² accuracy of 0.88 with an average prediction error of just 1.72 miles per gallon.

What Has Been Delivered

Component	Description
Trained ML Model	Random Forest Regressor trained on 392 vehicles. R ² = 0.88. Saved as model.pkl.
Flask Backend API	Python server exposing the model via HTTP. Accepts engine specs, returns predictions.
Frontend Dashboard	Premium dark-themed interactive dashboard with live predictions and fuel comparison. Built in pure HTML, CSS, and JavaScript.
5 Fuel Type Logic	Petrol, Diesel, Ethanol, CNG, Hydrogen — each with CO ₂ and efficiency modifiers.
CO ₂ Insights	Live business-relevant messaging tied to PHINIA's Net Zero values.
GitHub Repository	Full source code published to a public GitHub repository and available for immediate review.
Project Documentation	Full PDF documentation suite explaining build, architecture, and PHINIA relevance.

Headline Result: Switching a baseline engine from petrol to hydrogen — as predicted by this system — increases fuel efficiency by 35% and reduces direct CO₂ emissions from 120 g/km to 0 g/km. This single comparison is the entire commercial argument for PHINIA's hydrogen injection technology, quantified by an AI model.

02 — What This Project Is

The Fuel Efficiency Predictor is a working web application. A user opens the page, selects a fuel type — petrol, diesel, ethanol, CNG, or hydrogen — adjusts six sliders representing engine specifications, and clicks Run Prediction. Within milliseconds the system returns four outputs: predicted miles per gallon, CO₂ emissions in grams per kilometre, fuel consumption in litres per 100km, and a PHINIA-aligned insight message.

The Six Engine Inputs

Input	Description
Cylinders	Number of engine cylinders (3 to 8). Indicator of engine size.
Displacement	Engine volume in cubic inches (68 to 455). The strongest predictor of fuel use.
Horsepower	Engine power output (46 to 230 hp). Performance vs efficiency trade-off.
Weight	Vehicle weight in pounds (1500 to 5200). Heavier cars burn more fuel.
Acceleration	Time from 0 to 60 mph (8 to 25 seconds). Driving behaviour proxy.
Model Year	Manufacturing year (70 to 82). Captures technology improvements over time.

The Four Outputs

Output	Description
Predicted MPG	Miles per gallon prediction from the trained Random Forest model, adjusted by fuel multiplier.
CO ₂ Emissions	Direct CO ₂ output per kilometre based on fuel type. Hydrogen = 0, Petrol = 120, Diesel = 110, CNG = 55, Ethanol = 70.
Litres / 100km	European fuel consumption metric — automatically calculated from predicted MPG.
PHINIA Insight	Live business-relevant message comparing the selected fuel to petrol baseline and tying to PHINIA's strategy.

This is not a school exercise. This is a working product. Every input the user changes triggers a real machine learning prediction. Every result shown reflects what an engineer would see when designing a real engine configuration for an OEM customer.

03 — How It Was Built

The project was built in four sequential phases over approximately four weeks. Each phase produced a tangible deliverable that the next phase depended on.

Phase 1 — Environment Setup

Python 3.11 installed via pyenv. A dedicated virtual environment was created to isolate dependencies. Libraries installed: pandas, numpy, scikit-learn, matplotlib, seaborn, joblib, flask, flask-cors.

Tools used: Python 3.11, VS Code, Mac Terminal, Jupyter Notebook, Git.

Phase 2 — Data Acquisition and Cleaning

The UCI Auto MPG dataset was sourced from the University of California Irvine Machine Learning Repository. The raw file contained 398 vehicle records from 1970 to 1982 across 9 columns. Six rows with missing horsepower values were removed. The car_name column was dropped. The final cleaned dataset contained 392 complete rows with 8 numeric columns.

Phase 3 — Machine Learning Training

Three algorithms were trained and compared. The data was split 80% for training and 20% for testing. Each model was evaluated on R^2 score, Mean Absolute Error, and 5-fold cross-validation.

Algorithm	Performance
Linear Regression	R^2 : 0.79 · MAE: 2.42 mpg · Cross-Val: 0.59 · Not selected
Random Forest	R^2 : 0.88 · MAE: 1.72 mpg · Cross-Val: 0.76 · ✓ SELECTED
Gradient Boosting	R^2 : 0.87 · MAE: 1.80 mpg · Cross-Val: 0.75 · Not selected

Random Forest was selected as the production model. It was saved using joblib to model/model.pkl for use in the API.

Phase 4 — API and Frontend

A Flask backend was built to expose the trained model via HTTP. The /predict endpoint accepts a JSON payload of engine specifications, loads the saved model, applies the fuel type multiplier, calculates CO₂ emissions, and returns the result.

The frontend was built in pure HTML, CSS, and JavaScript — no frameworks. The design uses a premium dark aesthetic with animated mesh background, gradient cyan-to-purple

accents, and custom-styled sliders. The page calls the Flask API asynchronously using the `fetch()` API and updates result cards in real time.

04 — Project Components

The complete project consists of eight files organised into a clean directory structure, with full source code published to a public GitHub repository.

File	Function
auto-mpg.data	Raw dataset from UCI Machine Learning Repository — the source data the model learns from.
train_model.py	Python script that loads, cleans, splits, trains, evaluates and saves the model. Run once to produce model.pkl.
model.pkl	The saved trained Random Forest model. Loaded by app.py at runtime to generate predictions.
features.json	List of the 7 feature column names in the correct order. Ensures predictions use the right columns.
exploration.ipynb	Jupyter notebook used during development to explore data, generate visualisations, and validate hypotheses.
app.py	Flask backend API. Loads model.pkl, exposes /predict endpoint, applies fuel multipliers, calculates CO ₂ .
index.html	Frontend dashboard. Pure HTML/CSS/JS with premium design. Calls app.py via fetch() and renders results.
requirements.txt	List of all Python libraries needed to run the project. Generated with pip freeze.
README.md	Project description, installation instructions, and PHINIA relevance summary.

05 — How The System Works

When the user interacts with the dashboard, data flows through six distinct stages from the browser to the model and back. Below is a trace of a single prediction from click to result.

Step-by-Step Data Flow

Stage	Description
1. User Interaction	User selects a fuel type by clicking a pill (Petrol, Diesel, Ethanol, CNG or Hydrogen). User adjusts six sliders and clicks Run Prediction.
2. Browser Fetch	JavaScript reads the slider values and selected fuel. Builds a JSON payload and sends an HTTP POST request to the Flask API using the fetch() API.
3. Flask API	Flask server receives the POST request, extracts the JSON payload, builds a feature array in the correct order, and looks up the fuel multiplier and CO ₂ value.
4. ML Model	The trained Random Forest model predicts the base MPG from the feature array. The base MPG is multiplied by the fuel type's efficiency multiplier (e.g. hydrogen = 1.35, petrol = 1.00).
5. Response Build	Flask calculates litres per 100km from MPG ($235.21 \div \text{mpg}$). Builds a JSON response with mpg, co2, lper100km, and fuel_type. Returns it to the browser.
6. Results Rendered	JavaScript receives the response, updates the four result cards, selects an insight message based on CO ₂ level, and animates the CO ₂ comparison bars.

06 — Machine Learning Performance

The Random Forest model was rigorously evaluated across three metrics. This section explains what the results mean in engineering terms.

Accuracy Metrics

Metric	Meaning
R² Score: 0.88	The model explains 88% of the variation in fuel efficiency across the 392-vehicle dataset. Predictions are highly aligned with real-world results.
MAE: 1.72 mpg	On average, predictions are off by just 1.72 miles per gallon. For a car getting 28 mpg, the model predicts between 26.3 and 29.7 mpg.
Cross-Val R²: 0.76	When the data is split 5 different ways and retested each time, average accuracy is 76%. This confirms the model generalises well and is not overfitting.

Feature Importances

The model reveals which engine specifications are most predictive of fuel efficiency — critical insight for PHINIA’s engineering teams.

Feature	Importance
Displacement	41.6% — The strongest single predictor. Engine size directly affects fuel consumption.
Horsepower	16.9% — Power output is closely tied to fuel burn rate.
Weight	14.4% — Heavier vehicles require more fuel to move.
Cylinders	13.8% — Linked to engine size and power.
Model Year	10.5% — Captures technology improvements over time.
Acceleration	2.5% — Low impact due to standardised test conditions.
Origin	0.4% — Manufacturing region — almost irrelevant.

Critical insight for PHINIA: Engine displacement is the single biggest predictor of fuel efficiency at 41.6%. PHINIA’s core product — the fuel injector — directly controls how efficiently displacement is converted to power. The model has independently validated that PHINIA’s technology operates on the most impactful variable in fuel efficiency.

07 — Why This Benefits PHINIA

Every design decision in this project was made with PHINIA's specific strategic goals in mind. The following maps each component to a real PHINIA business priority.

Hydrogen and Alternative Fuels Strategy

PHINIA actively develops hydrogen fuel injectors, CNG systems, and ethanol delivery solutions. This project models exactly those five fuel types and quantifies their efficiency advantage over petrol and diesel baselines. When a customer asks how much better hydrogen is than petrol, this app provides the answer in real numbers — instantly.

Net Zero Mission

PHINIA's stated mission is to power a cleaner tomorrow. The CO₂ rating in this system shows 0 g/km for hydrogen versus 120 g/km for petrol. This single comparison — visible in the dashboard — is the entire story PHINIA tells every OEM customer, told in data.

Innovation Culture (250+ Patents Since 2023)

PHINIA values innovation. This project demonstrates how machine learning can accelerate fuel system R&D by predicting performance before any physical engine is tested. The R² of 0.88 proves AI can replace significant portions of early-stage prototype testing.

Global Scale (12,500 Employees, 40+ Locations)

PHINIA supplies fuel systems globally. This project's architecture — Flask API plus trained ML model — is production-grade and scalable. The same model that predicts one engine could process a fleet of thousands automatically.

Aerospace Expansion (2026)

PHINIA recently entered aerospace. Aerospace fuel systems require precision performance modelling. The displacement-based prediction logic in this model is directly applicable to aerospace propulsion where injection precision determines efficiency.

08 — Current Status and Next Steps

Completed

Component	Status
Data Pipeline	UCI dataset loaded, cleaned, and explored. 392 records ready for ML training.
Model Training	Three algorithms compared. Random Forest selected. $R^2 = 0.88$. Model saved to disk.
Backend API	Flask server with /predict endpoint. Loads model, applies fuel logic, returns JSON.
Frontend	Premium HTML/CSS/JS dashboard with live predictions, 5 fuel types, and CO ₂ insights.
PHINIA Logic	Custom CO ₂ multipliers and business-aligned insight messages for all fuel types.
GitHub Repository	Full source code pushed to a public GitHub repository. Available for review.
Documentation	Full project documentation suite delivered to PHINIA.

Version B — Planned Upgrade

The following capability is planned as Version B of this system — a significant upgrade to the current working product. It is not part of the current build but represents a clear and scoped next step.

Feature	Description
File Upload Layer	The ability to upload PDF, Word, or Excel files containing engine specification data. The system will extract the specs automatically and run batch predictions — enabling fleet-scale analysis without manual data entry.

Future Enhancements

Enhancement	Description
Modern Dataset	Replace UCI Auto MPG (1970–1982) with PHINIA's own current engine data for production-relevant predictions.
Hydrogen Modelling	Add specific injector performance variables that align with PHINIA's hydrogen technology.
Batch Pipeline	Automated data pipeline that processes thousands of vehicles in a single run.
Deeper ML	Experiment with neural networks, XGBoost, and ensemble stacking for further accuracy gains.

Live Telemetry	Connect to real-time engine telemetry data for live efficiency monitoring.
Live Deployment	Host the application on Vercel or Render so PHINIA can access the live app without installation.

The project is the evidence.
Of
Genuine interest in PHINIA's mission.

Built entirely as an unsolicited research exercise by a first-year Computer Science (AI) student — to demonstrate genuine interest in PHINIA's mission and the technical capability to contribute to it from day one.

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